Case Study 1: AI Based Heatwave Monitoring and Early Warning Group 2 Topic: Variability and Extreme Temperature Analysis Project: HeatSense

Overall Team Contributions

1. Advay Chavan 16015025012
Contribution : Developed core statistical analysis, distribution fitting, analytical synthesis, and the main execution pipeline. Integrated backend components and managed the main GitHub repository.
Role Performed: Project Lead and Backend Integrator.

2. Priyanka Keshava 16015025028
Contribution: Developed the frontend interface, implemented styling and interactivity, and integrated Chart.js for visualizing analytical results.
Role Performed: Frontend and UI/UX Lead.

3. Vinit Chaudhari 16015025011
Contribution: Implemented Monte Carlo simulations with 100,000 iterations, developed bootstrap confidence intervals, and cross-checked statistical results.
Role Performed: Simulation and Cross-Check Analyst.

4. Kanishak Desai 16015025013
Contribution: Collected and prepared climate data, worked on dataset creation, developed data ingestion scripts, and supported the climate data processing and forecast pipeline.
Role Performed: Data Engineering and Climate Ingestion Lead.

5. Ishika Asthana 16015025002
Contribution: Analyzed climatological trends, calculated decadal slopes, investigated temporal dependence, and worked on Markov transition matrices and autocorrelation analysis.
Role Performed: Climatological Trends and Temporal Clustering Analyst.

6. Mrunmai Bhagat 16015025005
Contribution: Prepared the theoretical foundations and methodology sections of the report, documented mathematical derivations, and conducted a literature survey.
Role Performed: Report Lead — Theoretical Foundations and Methodology.

7. Harsh Bhosle 16015025009
Contribution: Prepared the analytical synthesis sections of the report, documented project limitations, reviewed scientific consistency, and developed the project demonstration script.
Role Performed: Report Co-Lead — Analytical Synthesis and Scientific Audit.